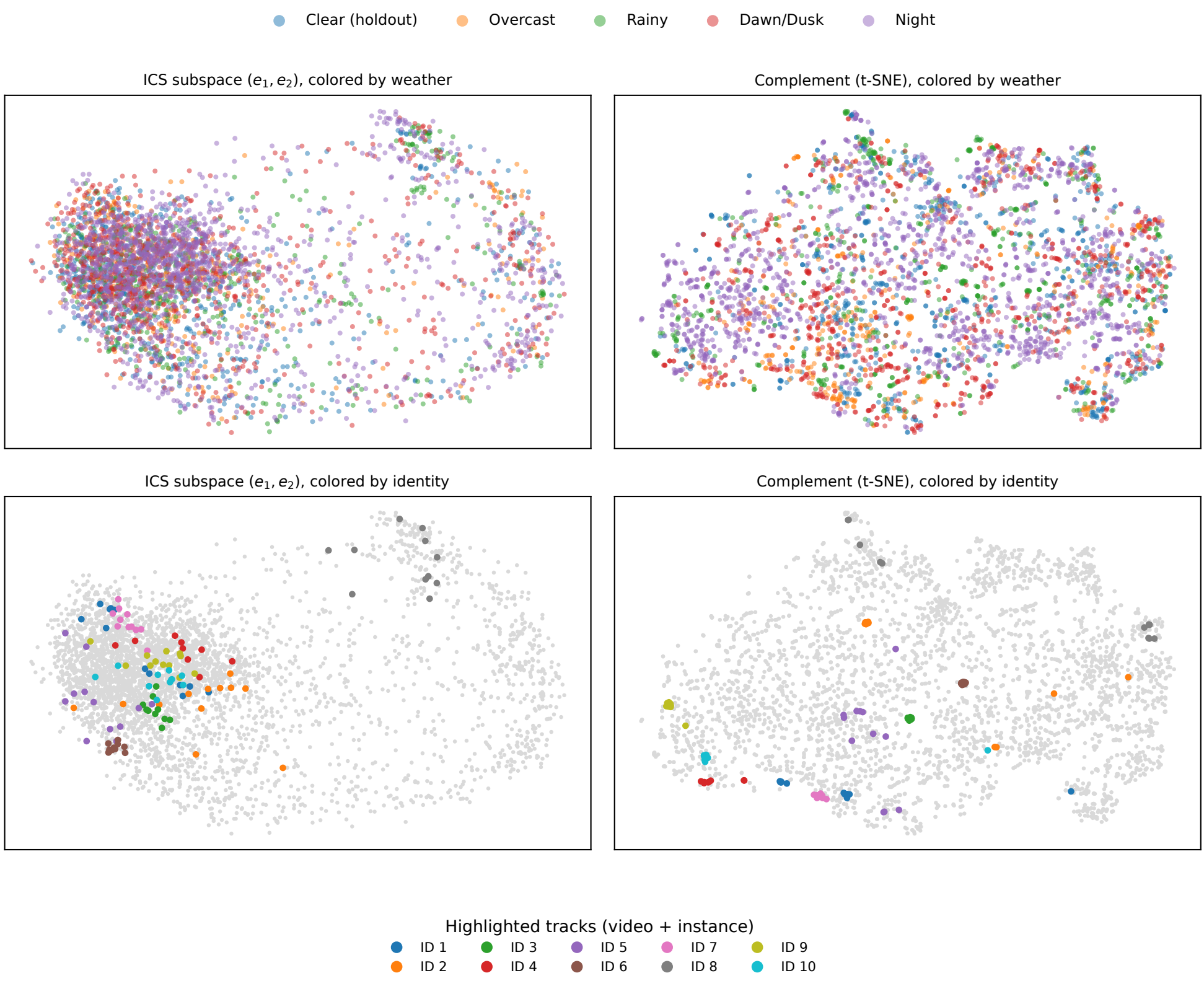
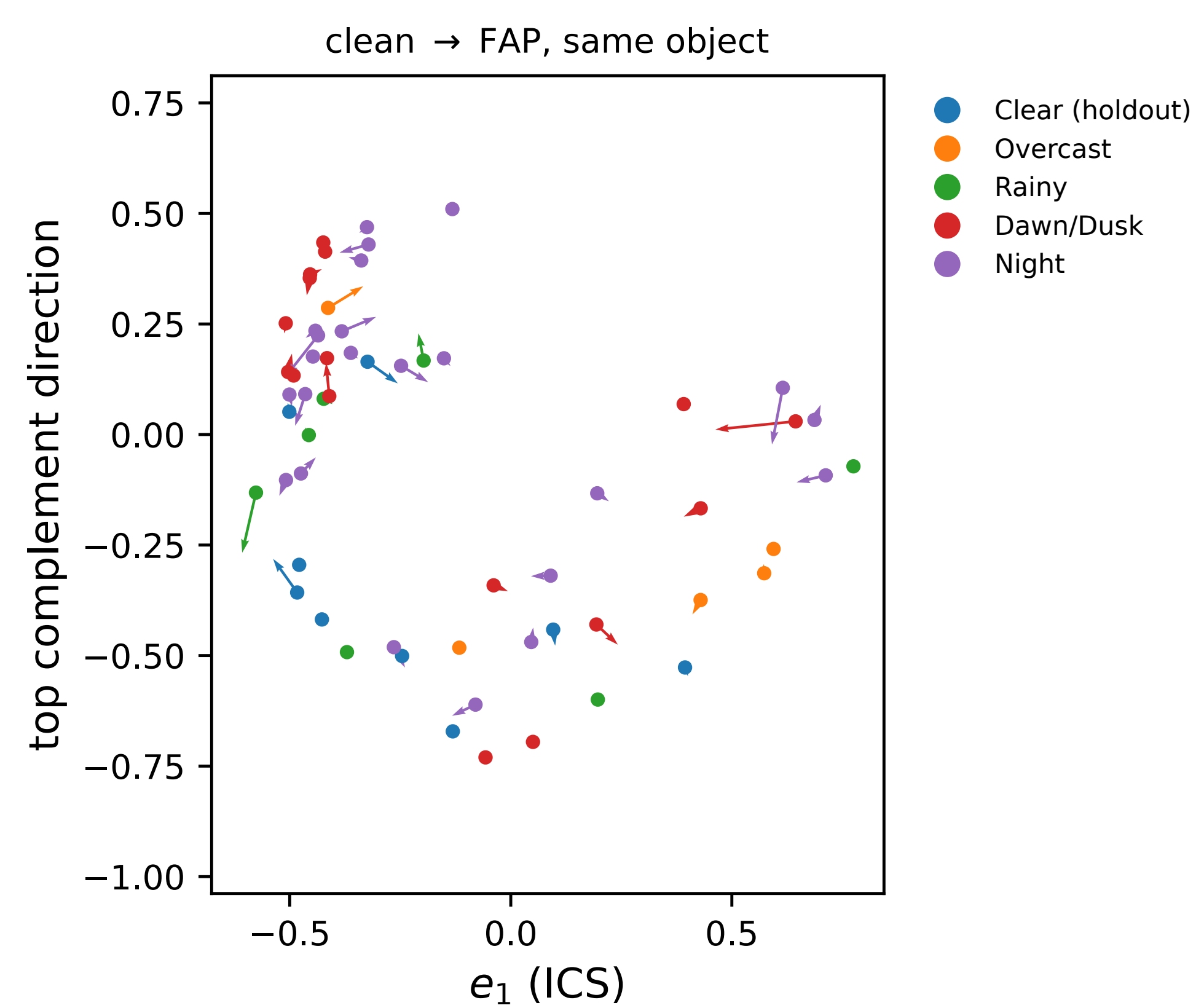
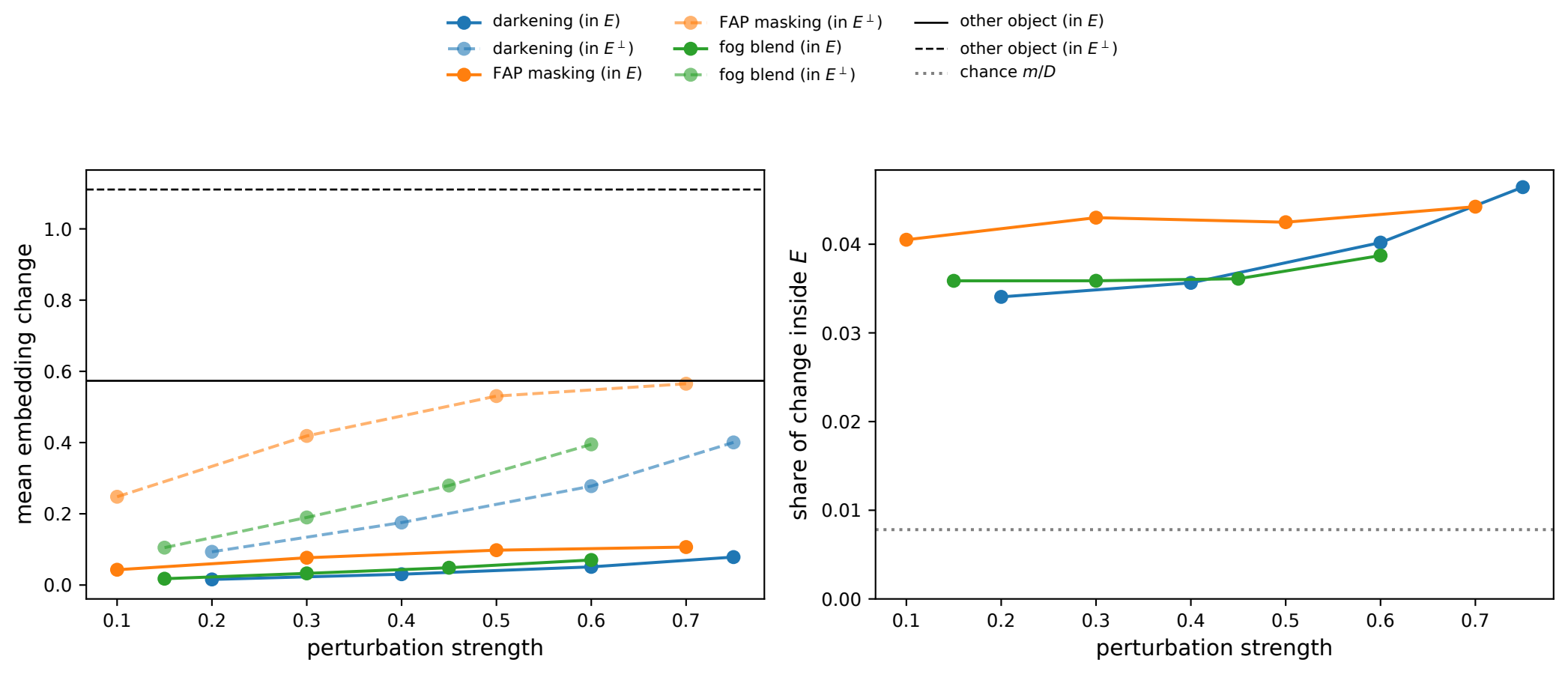
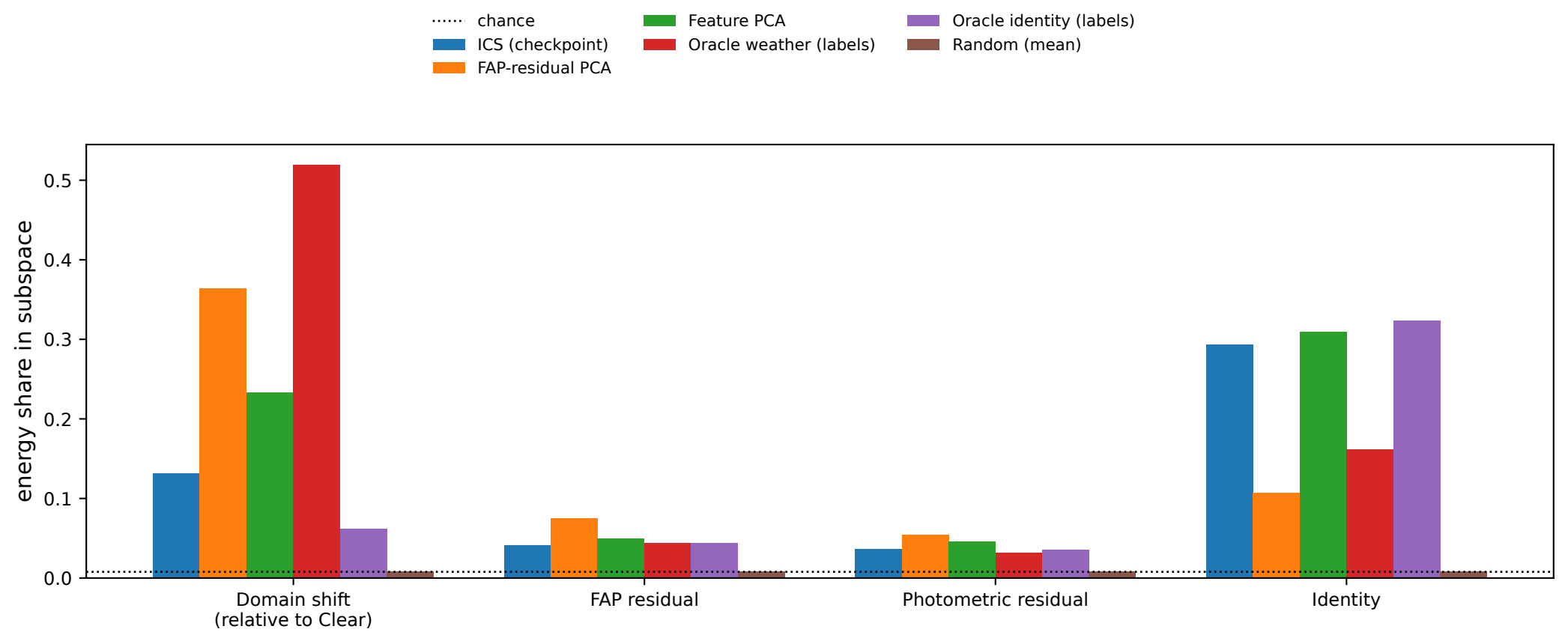

Clear (holdout)
Overcast
Rainy
Dawn/Dusk
Night
ICS subspace (e_1, e_2), colored by weather
Complement (t-SNE), colored by weather
ICS subspace (e_1, e_2), colored by identity
Complement (t-SNE), colored by identity
Highlighted tracks (video + instance)
ID 1
ID 2
ID 3
ID 4
ID 5
ID 6
ID 7
ID 8
ID 9
ID 10

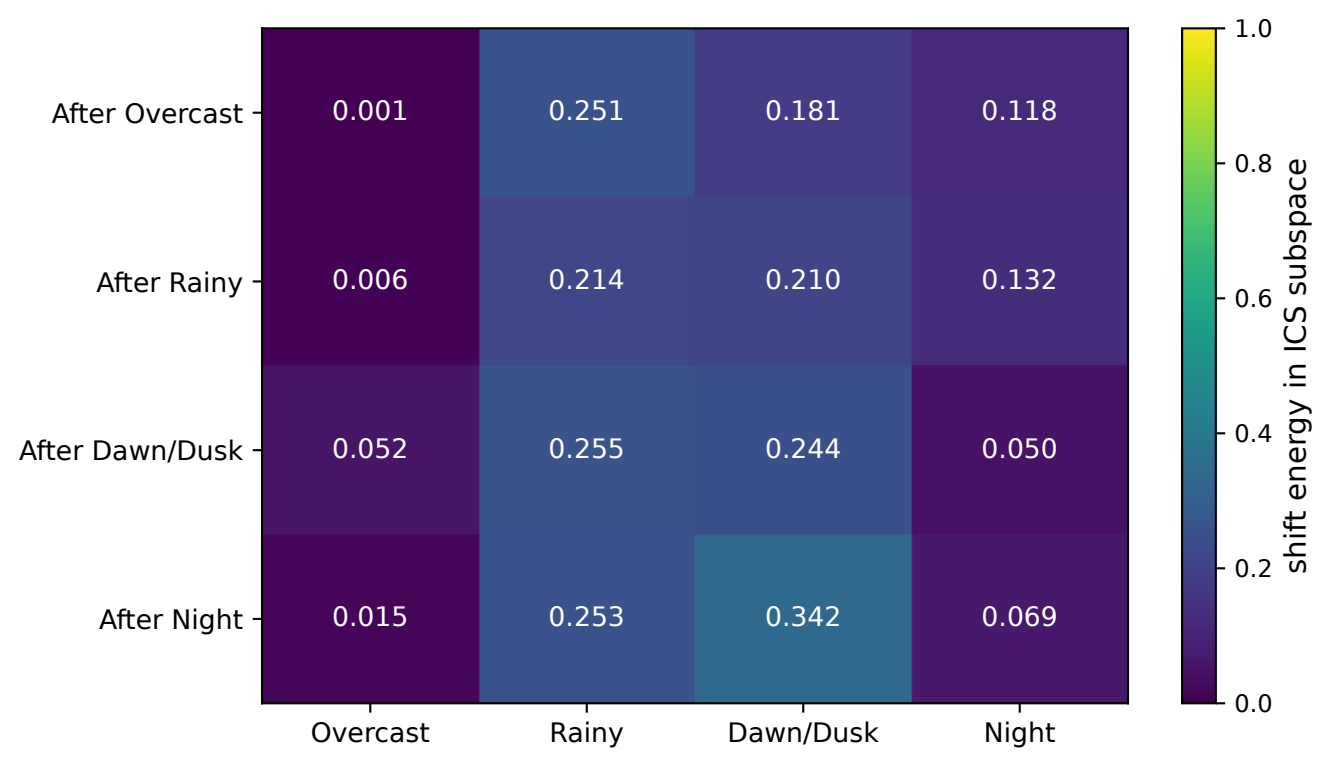

clean → FAP, same object
top complement direction
e_1 (ICS)
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−0.5
0.0
0.5
Clear (holdout)
Overcast
Rainy
Dawn/Dusk
Night


low e_1
Clear
-0.724
Night
-0.708
Dawn/Dusk
-0.705
Clear
-0.704
Overcast
-0.704
Clear
-0.701
Dawn/Dusk
-0.697
Dawn/Dusk
-0.694
Overcast
-0.692
Night
-0.688
high e_1
Night
+0.825
Dawn/Dusk
+0.816
Night
+0.788
Dawn/Dusk
+0.787
Dawn/Dusk
+0.787
Dawn/Dusk
+0.783
Clear
+0.778
Rainy
+0.775
Night
+0.774
Night
+0.774


low e_2
Night
-0.745
Dawn/Dusk
-0.735
Rainy
-0.731
Dawn/Dusk
-0.726
Night
-0.721
Dawn/Dusk
-0.713
Dawn/Dusk
-0.713
Dawn/Dusk
-0.711
Clear
-0.710
Clear
-0.709
high e_2
Night
+0.811
Night
+0.798
Night
+0.785
Night
+0.776
Dawn/Dusk
+0.775
Night
+0.774
Dawn/Dusk
+0.769
Dawn/Dusk
+0.763
Night
+0.758
Clear
+0.756


darkening (in E)
darkening (in $E^{\perp}$)
FAP masking (in E)
FAP masking (in $E^{\perp}$)
fog blend (in E)
fog blend (in $E^{\perp}$)
other object (in E)
other object (in $E^{\perp}$)
chance m/D
mean embedding change
perturbation strength
share of change inside E
perturbation strength


chance
ICS (checkpoint)
FAP-residual PCA
Feature PCA
Oracle weather (labels)
Oracle identity (labels)
Random (mean)
energy share in subspace
0.0
0.1
0.2
0.3
0.4
0.5
Domain shift
(relative to Clear)
FAP residual
Photometric residual
Identity


After Overcast
After Rainy
After Dawn/Dusk
After Night
0.001
0.251
0.181
0.118
0.006
0.214
0.210
0.132
0.052
0.255
0.244
0.050
0.015
0.253
0.342
0.069
Overcast
Rainy
Dawn/Dusk
Night
shift energy in ICS subspace
1.0
0.8
0.6
0.4
0.2
0.0